

Cinephone

Total Cost of Ownership Presentation

Presentation was based on previous project idea [SENTINEL].

OPERATOR

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STUDENT ID

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UNIT CODE

ICT171

HOST STATUS

ACTIVE

What is the project?



SENTINEL is a blog exploring how autonomous robotic systems could reshape physical security worldwide. Most security today relies on human guards, fixed cameras, and reactive alarms, all limited by fatigue, cost, and slow response times.



The blog examines how mobile autonomous units, equipped with sensors, computer vision, and decision-making logic, could patrol perimeters and detect anomalies faster and more consistently than existing systems.



Posts cover hardware, computer vision, coordination networks, and the ethics of machines making security decisions, using real-world pilots by companies and governments as case studies.



The site is hosted on a cloud VM running Apache, using Microsoft Azure.

LIVE PROJECT URL

cinemachine.sytes.net

PLATFORM

Microsoft Azure IaaS

Microsoft Azure

Ubuntu 22.04 · Apache · B2ats v2 VM (2 vCPU, 1 GiB RAM)

COST ITEM	YR 1	YR 2
VM Compute (B2ats v2, ~730h/mo) [1]	\$0 free	~\$215
Storage — 30 GB standard SSD [1]	~\$5	~\$5
Outbound bandwidth ~100 GB/mo [1]	~\$12	~\$12
Domain name (DDNS / custom) [2]	~\$15	~\$15
TLS— Let's Encrypt [2]	\$0 free	\$0 free
IT setup & maintenance (~10h @ \$30)	~\$300	~\$150
TOTAL (approx.)	~\$332	~\$397

YEAR 1

~\$332

Free compute tier applies

YEAR 2

~\$397

After free tier expires

Compute · Storage · Bandwidth · Domain · Labour

YEAR 1

~\$660

Higher upfront subscription cost

YEAR 2

~\$615

Slight saving as free domain expires

Subscription · Domain · Plugins · Labour

Option B

WordPress.com

Business Plan · Fully managed · No server configuration required

COST ITEM	YR 1	YR 2
WordPress.com Business Plan [3]	~\$540	~\$540
Custom domain name	\$0 free	~\$15
SSL certificate	\$0 free	\$0 free
Hosting mgmt & maintenance	\$0 free	\$0 free
Premium plugins/themes (est.)	~\$60	~\$30
IT setup time (~2h @ \$30)	~\$60	~\$30
TOTAL (approx.)	~\$660	~\$615

Why Azure



Transparency

With IaaS you can see exactly what's running, how it's configured. You trust the platform but you can't inspect it.



Full Control

Root access means any config, any software, any future expansion. SaaS platforms cap what you can touch.



Pay-As-You-Scale

Azure charges for what you use. A low-traffic blog stays cheap. SaaS charges a flat fee regardless.



Manual Maintenance

You manage OS patches, Apache updates, and security hardening yourself. This is a real time cost.



No Managed Failover

Azure does not provide automatic uptime guarantees on a basic VM. Downtime requires manual intervention.



You Own Security

Firewall rules, SSL renewal, intrusion detection, all your responsibility. SaaS handles this automatically.

The verdict

YEAR 1 COST COMPARISON

Azure

~\$332

WP.com

~\$660

YEAR 2 COST COMPARISON

Azure

~\$397

WP.com

~\$615

RECOMMENDED

Azure IaaS

Cheaper in both years.
Full infrastructure control.

References



- [1] Microsoft Corporation, "Azure Virtual Machines Pricing," Microsoft Azure, 2026. [Online]. Available: <https://azure.microsoft.com/en-au/pricing/details/virtual-machines/linux/> [Accessed: Jul. 2026].
- [2] Let's Encrypt, "About Let's Encrypt," Internet Security Research Group, 2026. [Online]. Available: <https://letsencrypt.org/about/> [Accessed: Jul. 2026].
- [3] Automattic, "WordPress.com Plans and Pricing," WordPress.com, 2026. [Online]. Available: <https://wordpress.com/pricing/> [Accessed: Jul. 2026].
- [4] No-IP, "Free Dynamic DNS," Vitalwerks Internet Solutions, 2026. [Online]. Available: <https://www.noip.com/> [Accessed: Jul. 2026].